

26 June P4 recall sheet

How to use this sheet

Recall the method before opening the answer. Labels distinguish what was reached in the lesson from official-reference completion and self-assessed gaps.

Method prompts

P4 Q1 - negative-index binomial [lesson hints; official reference completion]

- Put the expression into the form $\text{constant} \times (1+u)^n$ before expanding.
- For $(2+5x)^{-2}$, what are the first four terms and what condition must u satisfy?
- Answer: $1/4 - 5x/4 + 75x^2/16 - 125x^3/8 + \dots$, valid for $|x| < 2/5$.
- After multiplying by 16, $(4/(2+5x))^2 \approx 4 - 20x + 75x^2$.

P4 Q2 - volume of revolution [self-assessed gap; reference only]

- For rotation about the x -axis, use $V = \pi \int y^2 dx$.
- For $y = (5/3)\sqrt{x-1}$, $2 \leq x \leq 8$, evaluate $\pi \int_2^8 (25/9)(x-1) dx$.
- Answer: $200\pi/3$ cubic units.
- Scope reminder: $\pi \int y^2 dx$ is required P4 content; $\pi \int x^2 dy$ is not required for P4.

P4 Q4 - implicit differentiation [part (a) reached; part (b) reference completion]

- Differentiate $4x^2 + y^2 - 2xy = 24x$, keeping the full product-rule bracket.
- Answer: $dy/dx = (12 - 4x + y)/(y - x)$.
- At $P(a, b)$ with gradient 2, use both the derivative and the fact that P is on the curve: $2a + b = 12$, $4a^2 + b^2 - 2ab = 24a$.
- Official reference answer: $P = (2, 8)$; reject $(6, 0)$ because $b > 0$.

P4 Q5 - vector-line intersection [setup reached; reference completion]

- Equate each x -, y - and z -component of the two line equations.
- The lesson reached $\lambda = -9$, $\mu = -3$.
- Official reference: $\beta = -40$, $P = (-7, -37, -15)$.

P4 Q6 - trigonometric substitution [through power split; reference completion]

- Use $\sin 2\theta = 2\sin\theta\cos\theta$, $u=3+\cos\theta$, $du=-\sin\theta d\theta$, and $\cos\theta=u-3$.
- Transform limits in order: $\theta:0 \rightarrow \pi/2$ becomes $u:4 \rightarrow 3$.
- Reached form: $-2\int_4^3(u^{1/2}-3u^{-1/2})du$.
- Official reference: $8\sqrt{3}-40/3$, so $a=8$, $b=-40/3$.

Prior P3 review - range and inverse domain [not a P4 paper question]

- Rewrite $(3+5x)/(x+2)$ as $5-7/(x+2)$.
- With $x>-2$, the range is $y<5$; therefore the inverse domain is $x<5$.

P4 Q8 - separable differential equation [self-assessed gap; reference only]

- Integrate $dV/dt=12te^{-t}$ and apply $V(0)=6$.
- Answer: $V=18-12te^{-t}-12e^{-t}$.
- For $t\geq 0$, $V<18<20$, so capacity 20 is never reached.

Status key

- Reached: the furthest point actually worked live.
- Reference completion: official result retained for checking, not completed aloud.
- Self-assessed gap: named as a gap but not taught in the lesson.
- Prior P3 review: revisited material, not an October 2025 P4 question.